

TMM4250 Project work, Corotational nonlinear FEM code, v1

October 2026

The term project is performed by groups of 2 or 3 students. The students form the groups themselves. Term project grade accounts for 1/3 of the final grade in the course.

This programming part concerns the implementation of a nonlinear FE solver with 2D beam elements. The solver should have “Load control” and “Arc length” control algorithms. The two examples described below should be solved.

Problem 1: Nonlinear analysis of cantilever with end moment

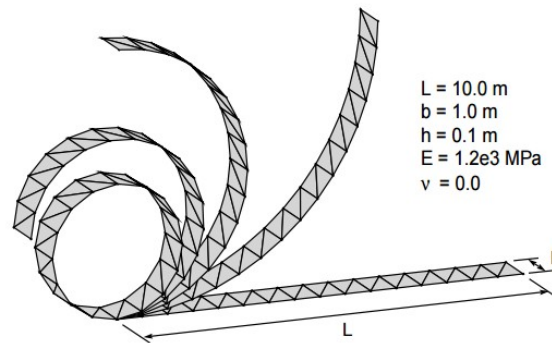


Figure 1: Example behaviour of a cantilever subject to end moment

1. Model a cantilever with end moment and perform linear analysis.
 - The `BeamModels_with_TODO.py` already has this model.
 - The linear algorithm is already implemented in `SolverAlgorithms_with_TODO.py`
 - The main file `SolveModel_main.py` is already set up to run this.
2. Implement a nonlinear formulation of a beam element using a co-rotational formulation from the lecture notes. *Note that only material stiffness is necessary for this problem to converge*
Implement a load control algorithm with Newton iterations.
3. Perform nonlinear analysis and bend the beam into a full loop. (Several loops are also possible)

1 Problem 2: Nonlinear analysis of deep arch model

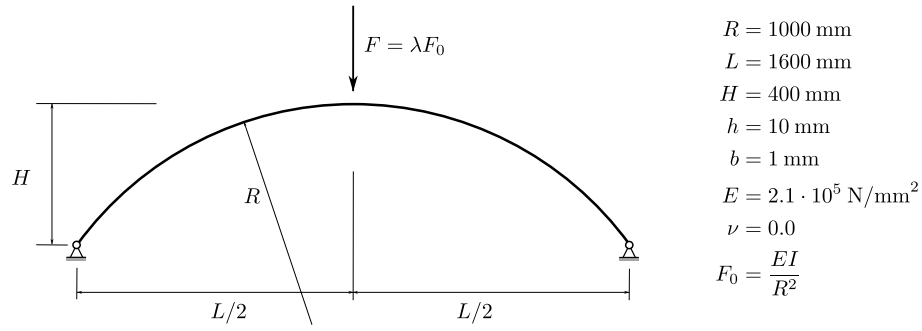


Figure 2: Deep arch geometry and material properties

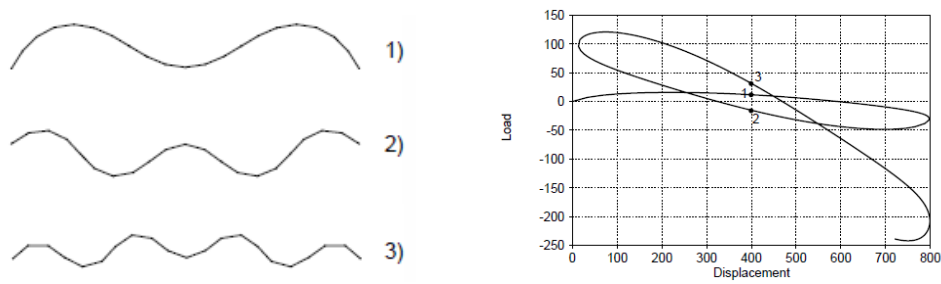


Figure 3: Deep arch displacements and load-deflection response

1. Model the deep arch model and perform a linear analysis.
2. Implement the geometric stiffness for the element if this was not done in the previous section.
3. Implement the arc-length algorithm described in the class notes as well as Section 3.2 of [1]
4. Perform nonlinear analysis for the arch

Document and report the code and results.

References

- [1] B Haugen. *Buckling and stability problems for thin shell structures using high performance finite elements*. PhD thesis, Norwegian University of Science and Technology, 1994.